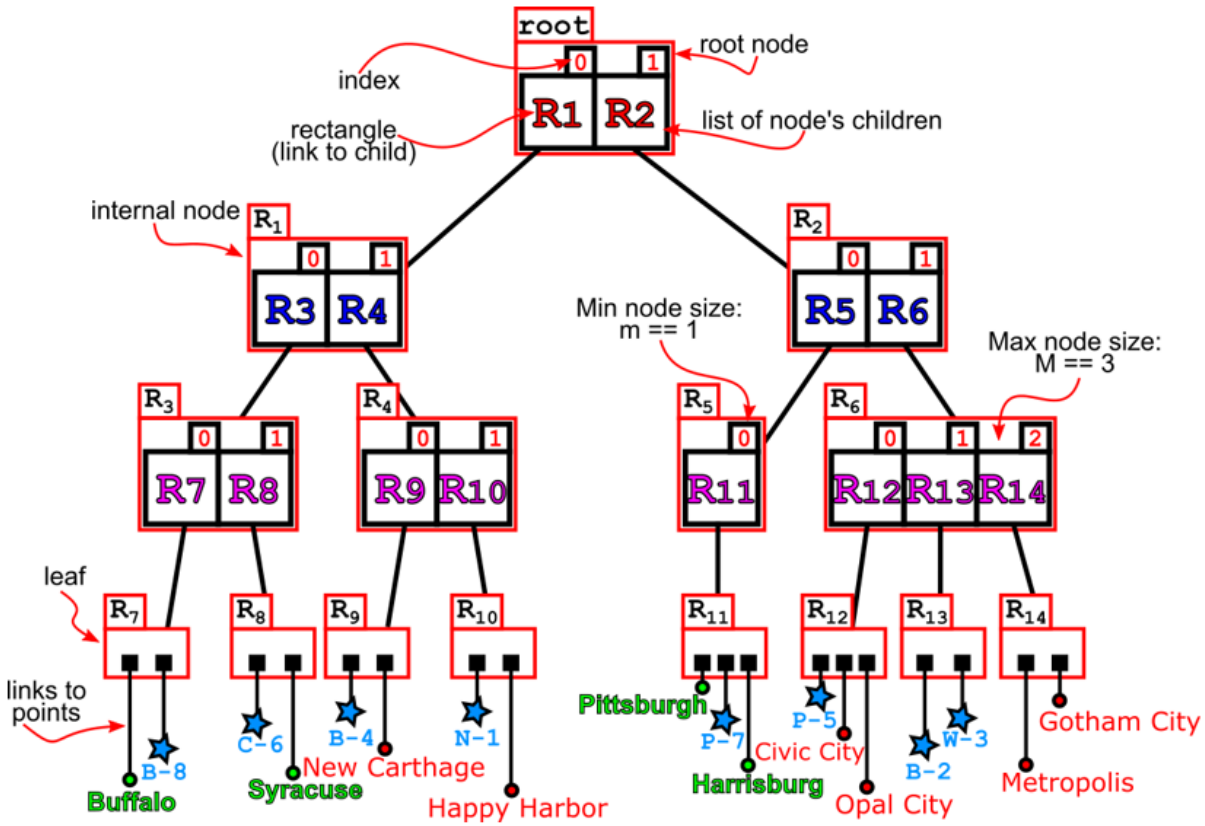


Figure 10.4 The tree representation for the R-tree of figure 10.3. The parameters for this R-tree are `m==1` and `M==3`. Internal nodes only hold bounding boxes, while leaves hold the actual points (or, in general, k-dimensional entries). In the rest of the chapter we will use a more compact representation, for each node drawing just the list of its children.

On removal, if any node becomes smaller than `m` entries, we will have to merge it with one of its adjacent siblings.

Invariants 2 and 4 require some extra work to be maintained true, but these bounding rectangles defined for each node are needed to allow fast search on the tree.